

Department of Mathematics, IIT Madras
MA 2031 - Linear Algebra for Engineers

Quiz I

Date: October 12, 2020. Time: 50 minutes Max. marks: 20

Answer all questions. Each question carries 4 marks.

1. Which of the following subsets of $\mathbb{R}^3$ are subspaces? Justify your conclusions.

(a) $W = \{(x, y, z) : x^3 = y^3\}$.

(b) $W = \{(x, y, z) : x^4 = y^4\}$.

2. Let A and B be subsets of a vector space V . Let “sp” denote the linear span. Show that

(a) $\text{sp}(A \cap B) \subseteq \text{sp}(A) \cap \text{sp}(B)$.

(b) Is the inclusion $\text{sp}(A) \cap \text{sp}(B) \subseteq \text{sp}(A \cap B)$ true? Justify your answer.

3. Determine the relationship between a, b if the following vectors in $\mathbb{R}^4$ are linearly *independent*.

$$\{(1, 1, 1, a), (1, 1, 0, 1), (-1, 2, b, -2), (1, 0, 0, 1)\}.$$

4. Suppose that subspaces U, W of $\mathbb{R}^3$ are defined by

$$U = \{(x, y, z) : x + y + 2z = 0\}, \quad W = \{(x, y, z) : x - y - 2z = 0\}.$$

Find bases for $U \cap W$ and $U + W$. Justify your answers.

5. Using Gram-Schmidt process, determine *all the vectors* that are orthogonal to the vector $(0, 1, 1)$, relative to the standard inner product in $\mathbb{R}^3$.
